

Exploratory_Analysis

2026-03-26

```
set.seed(1234)
data<-read.csv("~/Desktop/STAT4052/Final Project/D2D2016FoodStudy_data.csv")
str(data)
```

```
## 'data.frame':    2660 obs. of  55 variables:
## $ person_id      : int  10001 10001 10001 10001 10001 10001 10001 10001 10001 10002 10002 ...
## $ percent_discard: int   1 3 90 0 92 91 91 96 49 0 ...
## $ beef           : int   1 1 1 1 0 0 0 0 1 1 ...
## $ class1         : int   0 0 0 0 0 0 0 0 1 1 ...
## $ level2         : int   0 0 0 0 0 0 0 0 1 0 ...
## $ level3         : int   0 0 1 0 1 0 0 0 0 0 ...
## $ bestby        : int   0 0 0 1 1 0 0 1 0 0 ...
## $ useby         : int   0 0 1 0 0 0 0 0 1 1 ...
## $ date_days      : int   1 3 1 2 7 1 7 3 1 3 ...
## $ price         : num  4.79 12.79 4.79 6.39 2.49 ...
## $ large         : int   0 1 0 0 0 0 1 0 1 0 ...
## $ age           : num  49.5 49.5 49.5 49.5 49.5 49.5 49.5 49.5 69.5 69.5 ...
## $ gender        : int   0 0 0 0 0 0 0 0 0 0 ...
## $ educ          : int   0 0 0 0 0 0 0 0 0 0 ...
## $ income        : num  6.25 6.25 6.25 6.25 6.25 6.25 6.25 6.25 4.25 4.25 ...
## $ hhld_size     : int   2 2 2 2 2 2 2 2 1 1 ...
## $ child_pres    : int   0 0 0 0 0 0 0 0 0 0 ...
## $ nonwhite      : int   0 0 0 0 0 0 0 0 0 0 ...
## $ preshop_1     : int   4 4 4 4 4 4 4 4 1 1 ...
## $ preshop_2     : int   4 4 4 4 4 4 4 4 1 1 ...
## $ preshop_3     : int   4 4 4 4 4 4 4 4 1 1 ...
## $ preshop_4     : int   4 4 4 4 4 4 4 4 2 2 ...
## $ instore_5     : int   4 4 4 4 4 4 4 4 1 1 ...
## $ instore_6     : int   4 4 4 4 4 4 4 4 3 3 ...
## $ instore_7     : int   4 4 4 4 4 4 4 4 3 3 ...
## $ tosspre_8     : int   5 5 5 5 5 5 5 5 2 2 ...
## $ tosspre_9     : int   5 5 5 5 5 5 5 5 1 1 ...
## $ tosspre_10    : int   5 5 5 5 5 5 5 5 5 5 ...
## $ tosspre_11    : int   5 5 5 5 5 5 5 5 3 3 ...
## $ tosspre_12    : int   5 5 5 5 5 5 5 5 5 5 ...
## $ tosspre_13    : int   5 5 5 5 5 5 5 5 5 5 ...
## $ tosspre_14    : int   2 2 2 2 2 2 2 2 4 4 ...
## $ tosspost_15   : int   3 3 3 3 3 3 3 3 4 4 ...
## $ tosspost_16   : int   3 3 3 3 3 3 3 3 2 2 ...
## $ tosspost_17   : int   4 4 4 4 4 4 4 4 4 4 ...
## $ tosspost_18   : int   2 2 2 2 2 2 2 2 5 5 ...
## $ tosspost_19   : int   2 2 2 2 2 2 2 2 1 1 ...
## $ amount_20     : int   2 2 2 2 2 2 2 2 2 2 ...
## $ amount_21     : int   2 2 2 2 2 2 2 2 2 2 ...
```

```
## $ amount_22      : int  2 2 2 2 2 2 2 2 3 3 ...
## $ amount_23      : int  2 2 2 2 2 2 2 2 0 0 ...
## $ amount_24      : int  0 0 0 0 0 0 0 0 1 1 ...
## $ amount_25      : int  2 2 2 2 2 2 2 2 1 1 ...
## $ amount_26      : int  1 1 1 1 1 1 1 1 2 2 ...
## $ amount_27      : int  1 1 1 1 1 1 1 1 2 2 ...
## $ skill_28        : int  5 5 5 5 5 5 5 5 4 4 ...
## $ skill_29        : int  5 5 5 5 5 5 5 5 3 3 ...
## $ skill_30        : int  5 5 5 5 5 5 5 5 5 5 ...
## $ skill_31        : int  5 5 5 5 5 5 5 5 4 4 ...
## $ skill_32        : int  5 5 5 5 5 5 5 5 5 5 ...
## $ skill_33        : int  5 5 5 5 5 5 5 5 5 5 ...
## $ family_34       : int  2 2 2 2 2 2 2 2 4 4 ...
## $ family_35       : int  5 5 5 5 5 5 5 5 4 4 ...
## $ compost_36      : int  5 5 5 5 5 5 5 5 5 5 ...
## $ recycle_37      : int  5 5 5 5 5 5 5 5 1 1 ...
```

```
head(data)
```

```
##   person_id percent_discard beef class1 level2 level3 bestby useby date_days
## 1    10001             1     1     0     0     0     0     0         1
## 2    10001             3     1     0     0     0     0     0         3
## 3    10001            90     1     0     0     1     0     1         1
## 4    10001             0     1     0     0     0     1     0         2
## 5    10001            92     0     0     0     1     1     0         7
## 6    10001            91     0     0     0     0     0     0         1
##   price large  age gender educ income hhld_size child_pres nonwhite preshop_1
## 1  4.79     0 49.5     0   0   6.25         2         0         0         4
## 2 12.79     1 49.5     0   0   6.25         2         0         0         4
## 3  4.79     0 49.5     0   0   6.25         2         0         0         4
## 4  6.39     0 49.5     0   0   6.25         2         0         0         4
## 5  2.49     0 49.5     0   0   6.25         2         0         0         4
## 6  1.99     0 49.5     0   0   6.25         2         0         0         4
##   preshop_2 preshop_3 preshop_4 instore_5 instore_6 instore_7 tosspre_8
## 1         4         4         4         4         4         4         5
## 2         4         4         4         4         4         4         5
## 3         4         4         4         4         4         4         5
## 4         4         4         4         4         4         4         5
## 5         4         4         4         4         4         4         5
## 6         4         4         4         4         4         4         5
##   tosspre_9 tosspre_10 tosspre_11 tosspre_12 tosspre_13 tosspre_14 tosspost_15
## 1         5         5         5         5         5         2         3
## 2         5         5         5         5         5         2         3
## 3         5         5         5         5         5         2         3
## 4         5         5         5         5         5         2         3
## 5         5         5         5         5         5         2         3
## 6         5         5         5         5         5         2         3
##   tosspost_16 tosspost_17 tosspost_18 tosspost_19 amount_20 amount_21 amount_22
## 1           3           4           2           2           2           2         2
## 2           3           4           2           2           2           2         2
## 3           3           4           2           2           2           2         2
## 4           3           4           2           2           2           2         2
## 5           3           4           2           2           2           2         2
## 6           3           4           2           2           2           2         2
```

```
## amount_23 amount_24 amount_25 amount_26 amount_27 skill_28 skill_29 skill_30
## 1      2      0      2      1      1      5      5      5
## 2      2      0      2      1      1      5      5      5
## 3      2      0      2      1      1      5      5      5
## 4      2      0      2      1      1      5      5      5
## 5      2      0      2      1      1      5      5      5
## 6      2      0      2      1      1      5      5      5
## skill_31 skill_32 skill_33 family_34 family_35 compost_36 recycle_37
## 1      5      5      5      2      5      5      5
## 2      5      5      5      2      5      5      5
## 3      5      5      5      2      5      5      5
## 4      5      5      5      2      5      5      5
## 5      5      5      5      2      5      5      5
## 6      5      5      5      2      5      5      5
```

```
summary(data)
```

```
## person_id percent_discard beef class1
## Min. :10001 Min. : 0.00 Min. :0.0000 Min. :0.0000
## 1st Qu.:10084 1st Qu.: 0.00 1st Qu.:0.0000 1st Qu.:0.0000
## Median :10167 Median : 9.00 Median :0.0000 Median :0.0000
## Mean :10167 Mean : 27.02 Mean :0.4429 Mean :0.4323
## 3rd Qu.:10250 3rd Qu.: 50.00 3rd Qu.:1.0000 3rd Qu.:1.0000
## Max. :10333 Max. :100.00 Max. :1.0000 Max. :1.0000
##
## level2 level3 bestby useby
## Min. :0.0000 Min. :0.000 Min. :0.0000 Min. :0.0000
## 1st Qu.:0.0000 1st Qu.:0.000 1st Qu.:0.0000 1st Qu.:0.0000
## Median :0.0000 Median :0.000 Median :0.0000 Median :0.0000
## Mean :0.2462 Mean :0.244 Mean :0.3391 Mean :0.3289
## 3rd Qu.:0.0000 3rd Qu.:0.000 3rd Qu.:1.0000 3rd Qu.:1.0000
## Max. :1.0000 Max. :1.000 Max. :1.0000 Max. :1.0000
##
## date_days price large age
## Min. :1.000 Min. : 1.990 Min. :0.0000 Min. :21.00
## 1st Qu.:1.000 1st Qu.: 2.990 1st Qu.:0.0000 1st Qu.:29.50
## Median :3.000 Median : 4.990 Median :1.0000 Median :49.50
## Mean :2.926 Mean : 6.341 Mean :0.5071 Mean :45.41
## 3rd Qu.:3.000 3rd Qu.: 7.990 3rd Qu.:1.0000 3rd Qu.:59.50
## Max. :7.000 Max. :15.990 Max. :1.0000 Max. :84.50
##
## gender educ income hhld_size
## Min. :0.0000 Min. :0.0000 Min. : 1.850 Min. :1.00
## 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.: 4.250 1st Qu.:2.00
## Median :0.0000 Median :1.0000 Median : 8.750 Median :2.00
## Mean :0.3274 Mean :0.6669 Mean : 9.159 Mean :2.64
## 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:12.500 3rd Qu.:4.00
## Max. :1.0000 Max. :1.0000 Max. :25.000 Max. :7.00
##
## child_pres nonwhite preshop_1 preshop_2
## Min. :0.0000 Min. :0.0000 Min. :1.000 Min. :1.000
## 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:2.000 1st Qu.:2.000
## Median :0.0000 Median :0.0000 Median :3.000 Median :4.000
## Mean :0.2406 Mean :0.1414 Mean :2.956 Mean :3.342
```

```

## 3rd Qu.:0.0000 3rd Qu.:0.0000 3rd Qu.:4.000 3rd Qu.:4.000
## Max. :1.0000 Max. :1.0000 Max. :5.000 Max. :5.000
##
## preshop_3 preshop_4 instore_5 instore_6
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:3.000 1st Qu.:4.000 1st Qu.:3.000 1st Qu.:2.000
## Median :4.000 Median :4.000 Median :4.000 Median :3.000
## Mean :3.845 Mean :4.028 Mean :3.524 Mean :3.215
## 3rd Qu.:5.000 3rd Qu.:5.000 3rd Qu.:4.000 3rd Qu.:4.000
## Max. :5.000 Max. :5.000 Max. :5.000 Max. :5.000
##
## instore_7 tosspre_8 tosspre_9 tosspre_10
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:3.000 1st Qu.:2.000 1st Qu.:2.000 1st Qu.:4.000
## Median :4.000 Median :4.000 Median :3.000 Median :4.000
## Mean :3.582 Mean :3.398 Mean :3.114 Mean :4.138
## 3rd Qu.:4.000 3rd Qu.:4.000 3rd Qu.:4.000 3rd Qu.:5.000
## Max. :5.000 Max. :5.000 Max. :5.000 Max. :5.000
## NA's :47
## tosspre_11 tosspre_12 tosspre_13 tosspre_14
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:2.000 1st Qu.:2.000 1st Qu.:2.000 1st Qu.:1.000
## Median :2.000 Median :4.000 Median :2.000 Median :1.000
## Mean :2.747 Mean :3.492 Mean :2.816 Mean :1.701
## 3rd Qu.:4.000 3rd Qu.:5.000 3rd Qu.:4.000 3rd Qu.:2.000
## Max. :5.000 Max. :5.000 Max. :5.000 Max. :5.000
##
## tosspost_15 tosspost_16 tosspost_17 tosspost_18 tosspost_19
## Min. :1.000 Min. :1.00 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:2.000 1st Qu.:2.00 1st Qu.:2.000 1st Qu.:3.000 1st Qu.:1.000
## Median :4.000 Median :3.00 Median :4.000 Median :4.000 Median :2.000
## Mean :3.337 Mean :3.33 Mean :3.547 Mean :3.842 Mean :2.065
## 3rd Qu.:5.000 3rd Qu.:4.00 3rd Qu.:5.000 3rd Qu.:5.000 3rd Qu.:2.000
## Max. :5.000 Max. :5.00 Max. :5.000 Max. :5.000 Max. :5.000
##
## amount_20 amount_21 amount_22 amount_23
## Min. :0.000 Min. :0.000 Min. :0.000 Min. :0.000
## 1st Qu.:2.000 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000
## Median :2.000 Median :2.000 Median :1.000 Median :2.000
## Mean :2.457 Mean :1.715 Mean :1.676 Mean :1.956
## 3rd Qu.:3.000 3rd Qu.:2.000 3rd Qu.:2.000 3rd Qu.:3.000
## Max. :5.000 Max. :5.000 Max. :5.000 Max. :5.000
##
## amount_24 amount_25 amount_26 amount_27
## Min. :0.000 Min. :0.000 Min. :0.000 Min. :0.000
## 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000
## Median :2.000 Median :1.000 Median :1.000 Median :1.000
## Mean :1.935 Mean :1.496 Mean :1.667 Mean :1.316
## 3rd Qu.:3.000 3rd Qu.:2.000 3rd Qu.:2.000 3rd Qu.:2.000
## Max. :5.000 Max. :5.000 Max. :5.000 Max. :5.000
##
## skill_28 skill_29 skill_30 skill_31
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:4.000 1st Qu.:3.000 1st Qu.:4.000 1st Qu.:4.000

```

```
## Median :4.000 Median :4.000 Median :4.000 Median :4.000
## Mean :4.127 Mean :3.794 Mean :4.055 Mean :4.053
## 3rd Qu.:5.000 3rd Qu.:5.000 3rd Qu.:5.000 3rd Qu.:5.000
## Max. :5.000 Max. :5.000 Max. :5.000 Max. :5.000
##
## skill_32 skill_33 family_34 family_35 compost_36
## Min. :2.000 Min. :2.00 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:4.000 1st Qu.:4.00 1st Qu.:3.000 1st Qu.:3.000 1st Qu.:4.000
## Median :4.000 Median :4.00 Median :4.000 Median :4.000 Median :5.000
## Mean :4.108 Mean :4.27 Mean :3.423 Mean :3.707 Mean :4.366
## 3rd Qu.:5.000 3rd Qu.:5.00 3rd Qu.:4.000 3rd Qu.:4.000 3rd Qu.:5.000
## Max. :5.000 Max. :5.00 Max. :5.000 Max. :5.000 Max. :5.000
##
## recycle_37
## Min. :1.000
## 1st Qu.:1.000
## Median :1.000
## Mean :2.003
## 3rd Qu.:2.000
## Max. :5.000
##
```

```
dim(data)
```

```
## [1] 2660 55
```

```
data$beef<-as.factor(data$beef)
data$class1<-as.factor(data$class1)
data$level2<-as.factor(data$level2)
data$level3<-as.factor(data$level3)
data$bestby<-as.factor(data$bestby)
data$useby<-as.factor(data$useby)
data$large<-as.factor(data$large)
data$gender<-as.factor(data$gender)
data$educ<-as.factor(data$educ)
data$child_pres<-as.factor(data$child_pres)
data$nonwhite<-as.factor(data$nonwhite)
str(data)
```

```
## 'data.frame': 2660 obs. of 55 variables:
## $ person_id : int 10001 10001 10001 10001 10001 10001 10001 10001 10001 10002 10002 ...
## $ percent_discard: int 1 3 90 0 92 91 91 96 49 0 ...
## $ beef : Factor w/ 2 levels "0","1": 2 2 2 2 1 1 1 1 2 2 ...
## $ class1 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 2 ...
## $ level2 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
## $ level3 : Factor w/ 2 levels "0","1": 1 1 2 1 2 1 1 1 1 1 ...
## $ bestby : Factor w/ 2 levels "0","1": 1 1 1 2 2 1 1 2 1 1 ...
## $ useby : Factor w/ 2 levels "0","1": 1 1 2 1 1 1 1 1 2 2 ...
## $ date_days : int 1 3 1 2 7 1 7 3 1 3 ...
## $ price : num 4.79 12.79 4.79 6.39 2.49 ...
## $ large : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
## $ age : num 49.5 49.5 49.5 49.5 49.5 49.5 49.5 49.5 69.5 69.5 ...
## $ gender : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
```

```
## $ educ      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
## $ income    : num  6.25 6.25 6.25 6.25 6.25 6.25 6.25 6.25 4.25 4.25 ...
## $ hhld_size : int   2 2 2 2 2 2 2 2 1 1 ...
## $ child_pres : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
## $ nonwhite   : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
## $ preshop_1  : int   4 4 4 4 4 4 4 4 1 1 ...
## $ preshop_2  : int   4 4 4 4 4 4 4 4 1 1 ...
## $ preshop_3  : int   4 4 4 4 4 4 4 4 1 1 ...
## $ preshop_4  : int   4 4 4 4 4 4 4 4 2 2 ...
## $ instore_5  : int   4 4 4 4 4 4 4 4 1 1 ...
## $ instore_6  : int   4 4 4 4 4 4 4 4 3 3 ...
## $ instore_7  : int   4 4 4 4 4 4 4 4 3 3 ...
## $ tosspre_8  : int   5 5 5 5 5 5 5 5 2 2 ...
## $ tosspre_9  : int   5 5 5 5 5 5 5 5 1 1 ...
## $ tosspre_10 : int   5 5 5 5 5 5 5 5 5 5 ...
## $ tosspre_11 : int   5 5 5 5 5 5 5 5 3 3 ...
## $ tosspre_12 : int   5 5 5 5 5 5 5 5 5 5 ...
## $ tosspre_13 : int   5 5 5 5 5 5 5 5 5 5 ...
## $ tosspre_14 : int   2 2 2 2 2 2 2 2 4 4 ...
## $ tosspost_15 : int   3 3 3 3 3 3 3 3 4 4 ...
## $ tosspost_16 : int   3 3 3 3 3 3 3 3 2 2 ...
## $ tosspost_17 : int   4 4 4 4 4 4 4 4 4 4 ...
## $ tosspost_18 : int   2 2 2 2 2 2 2 2 5 5 ...
## $ tosspost_19 : int   2 2 2 2 2 2 2 2 1 1 ...
## $ amount_20  : int   2 2 2 2 2 2 2 2 2 2 ...
## $ amount_21  : int   2 2 2 2 2 2 2 2 2 2 ...
## $ amount_22  : int   2 2 2 2 2 2 2 2 3 3 ...
## $ amount_23  : int   2 2 2 2 2 2 2 2 0 0 ...
## $ amount_24  : int   0 0 0 0 0 0 0 0 1 1 ...
## $ amount_25  : int   2 2 2 2 2 2 2 2 1 1 ...
## $ amount_26  : int   1 1 1 1 1 1 1 1 2 2 ...
## $ amount_27  : int   1 1 1 1 1 1 1 1 2 2 ...
## $ skill_28   : int   5 5 5 5 5 5 5 5 4 4 ...
## $ skill_29   : int   5 5 5 5 5 5 5 5 3 3 ...
## $ skill_30   : int   5 5 5 5 5 5 5 5 5 5 ...
## $ skill_31   : int   5 5 5 5 5 5 5 5 4 4 ...
## $ skill_32   : int   5 5 5 5 5 5 5 5 5 5 ...
## $ skill_33   : int   5 5 5 5 5 5 5 5 5 5 ...
## $ family_34  : int   2 2 2 2 2 2 2 2 4 4 ...
## $ family_35  : int   5 5 5 5 5 5 5 5 4 4 ...
## $ compost_36 : int   5 5 5 5 5 5 5 5 5 5 ...
## $ recycle_37 : int   5 5 5 5 5 5 5 5 1 1 ...
```

```
data$beef<-factor(data$beef, levels=c(0,1),labels=c("Spinach", "Beef"))
data$class1<-factor(data$class1, levels = c(0,1),labels=c("Planner", "Extemporaneous"))
data$large<-factor(data$large, levels = c(0,1),labels=c("Small", "large"))
data$gender<-factor(data$gender, levels = c(0,1),labels=c("Not Female", "Female"))
```

```
data$educ <- factor(data$educ, levels = c(0,1),labels=c("No Degree", "Bachelors or More"))
data$child_pres <- factor(data$child_pres, levels = c(0,1),labels=c("No children", "has children"))
data$nonwhite<-factor(data$nonwhite, levels = c(0,1),labels = c("White", "Not White"))
colSums(is.na(data))
```

```
##      person_id percent_discard      beef      class1      level2
```

```
##          0          0          0          0          0
##      level3      bestby      useby      date_days      price
##          0          0          0          0          0
##      large      age      gender      educ      income
##          0          0          0          0          0
##      hhld_size      child_pres      nonwhite      preshop_1      preshop_2
##          0          0          0          0          0
##      preshop_3      preshop_4      instore_5      instore_6      instore_7
##          0          0          95          16          47
##      tosspre_8      tosspre_9      tosspre_10      tosspre_11      tosspre_12
##          0          0          0          0          0
##      tosspre_13      tosspre_14      tosspost_15      tosspost_16      tosspost_17
##          0          0          0          0          0
##      tosspost_18      tosspost_19      amount_20      amount_21      amount_22
##          0          0          0          0          0
##      amount_23      amount_24      amount_25      amount_26      amount_27
##          0          0          0          0          0
##      skill_28      skill_29      skill_30      skill_31      skill_32
##          0          0          0          0          0
##      skill_33      family_34      family_35      compost_36      recycle_37
##          0          0          0          0          0
```

```
data2 <- na.omit(data)
colSums(is.na(data2))
```

```
##      person_id percent_discard      beef      class1      level2
##          0          0          0          0          0
##      level3      bestby      useby      date_days      price
##          0          0          0          0          0
##      large      age      gender      educ      income
##          0          0          0          0          0
##      hhld_size      child_pres      nonwhite      preshop_1      preshop_2
##          0          0          0          0          0
##      preshop_3      preshop_4      instore_5      instore_6      instore_7
##          0          0          0          0          0
##      tosspre_8      tosspre_9      tosspre_10      tosspre_11      tosspre_12
##          0          0          0          0          0
##      tosspre_13      tosspre_14      tosspost_15      tosspost_16      tosspost_17
##          0          0          0          0          0
##      tosspost_18      tosspost_19      amount_20      amount_21      amount_22
##          0          0          0          0          0
##      amount_23      amount_24      amount_25      amount_26      amount_27
##          0          0          0          0          0
##      skill_28      skill_29      skill_30      skill_31      skill_32
##          0          0          0          0          0
##      skill_33      family_34      family_35      compost_36      recycle_37
##          0          0          0          0          0
```

```
summary(data2$percent_discard)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   0.00    9.00   26.92   50.00   100.00
```

```
summary(data2$price)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.990   2.990   4.990   6.337   7.990  15.990
```

```
summary(data2$age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    21.0   29.5   49.5   45.5   59.5   84.5
```

```
summary(data2$income)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.850   4.250   6.250   9.154  12.500  25.000
```

```
summary(data2$beef)
```

```
## Spinach    Beef
##    1414    1119
```

```
summary(data2$class1)
```

```
##      Planner Extemporaneous
##      1454          1079
```

```
summary(data2$gender)
```

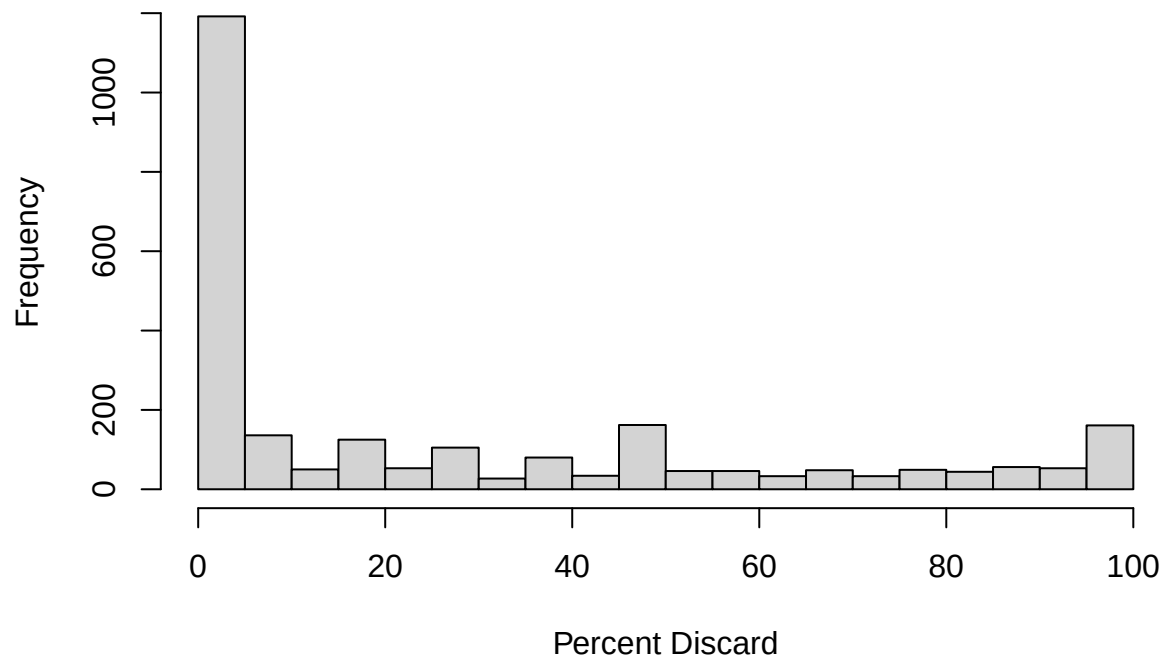
```
## Not Female    Female
##      1742      791
```

```
summary(data2$educ)
```

```
##      No Degree Bachelors or More
##      839          1694
```

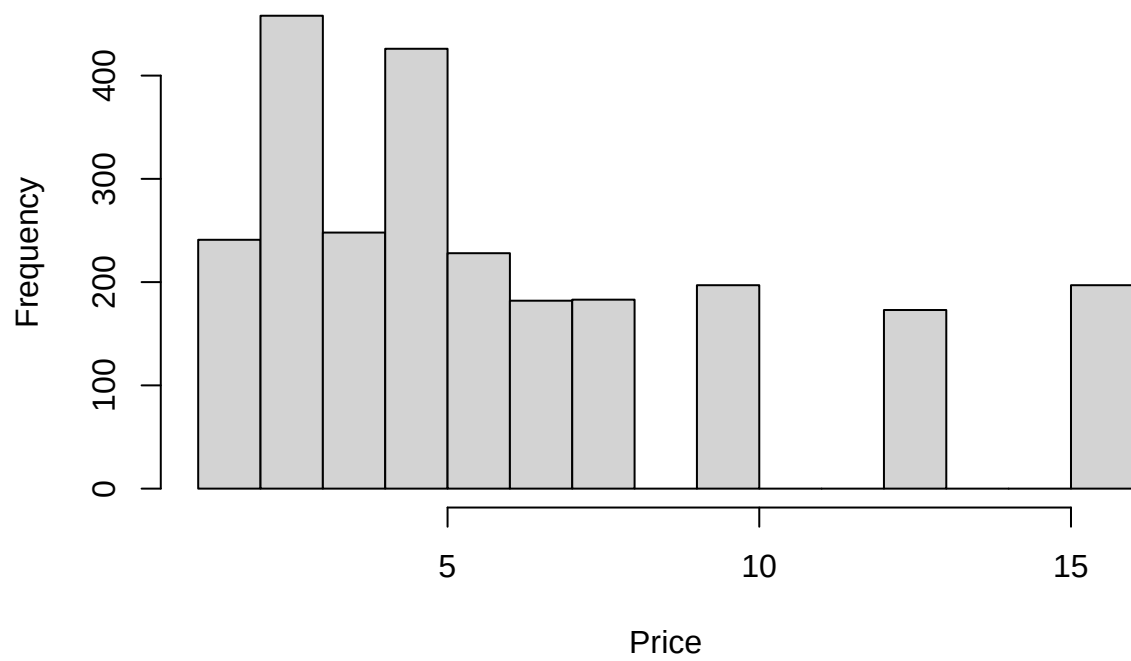
```
hist(data2$percent_discard,main="Percent Discard", xlab="Percent Discard", breaks =30)
```

Percent Discard

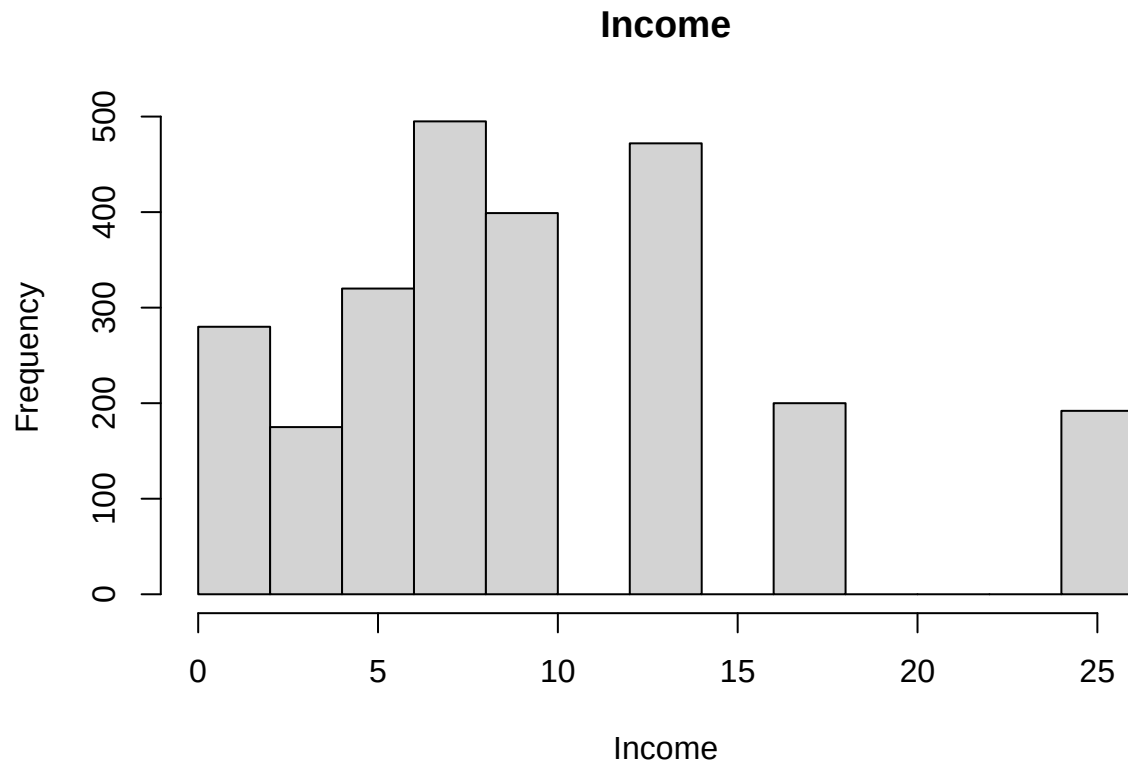


```
hist(data2$price,main="Price",xlab="Price")
```

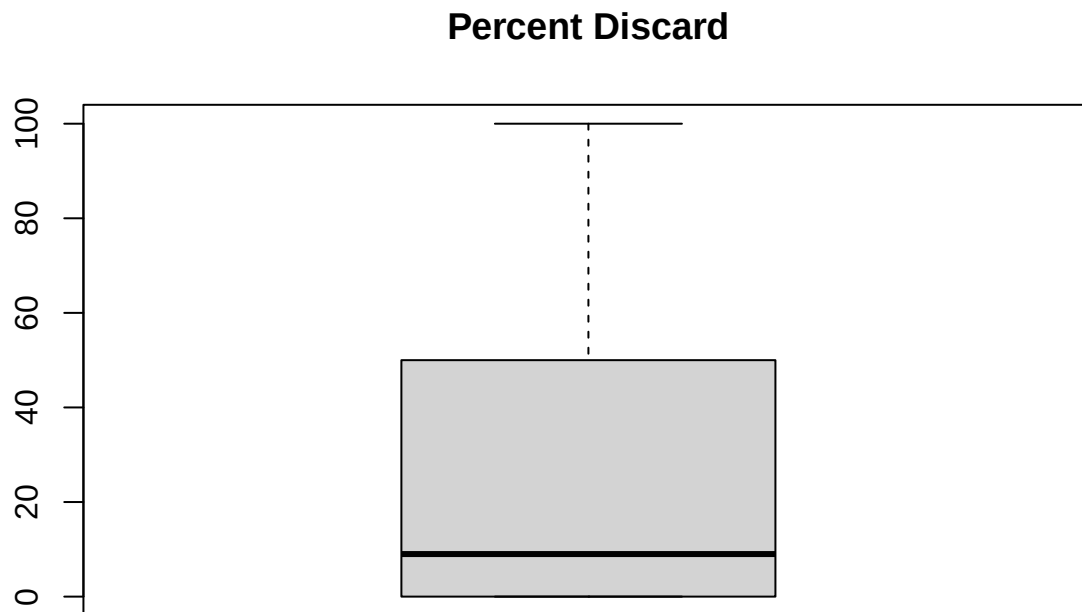
Price



```
hist(data2$income,main="Income",xlab="Income")
```

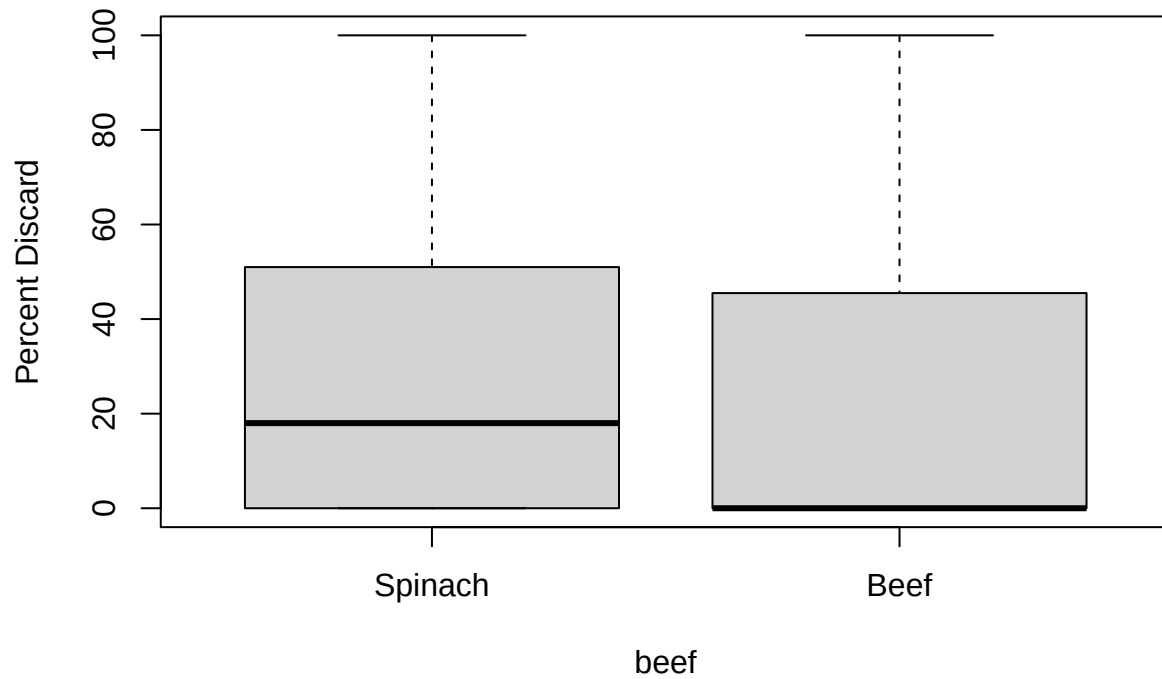


```
boxplot(data2$percent_discard,main="Percent Discard")
```



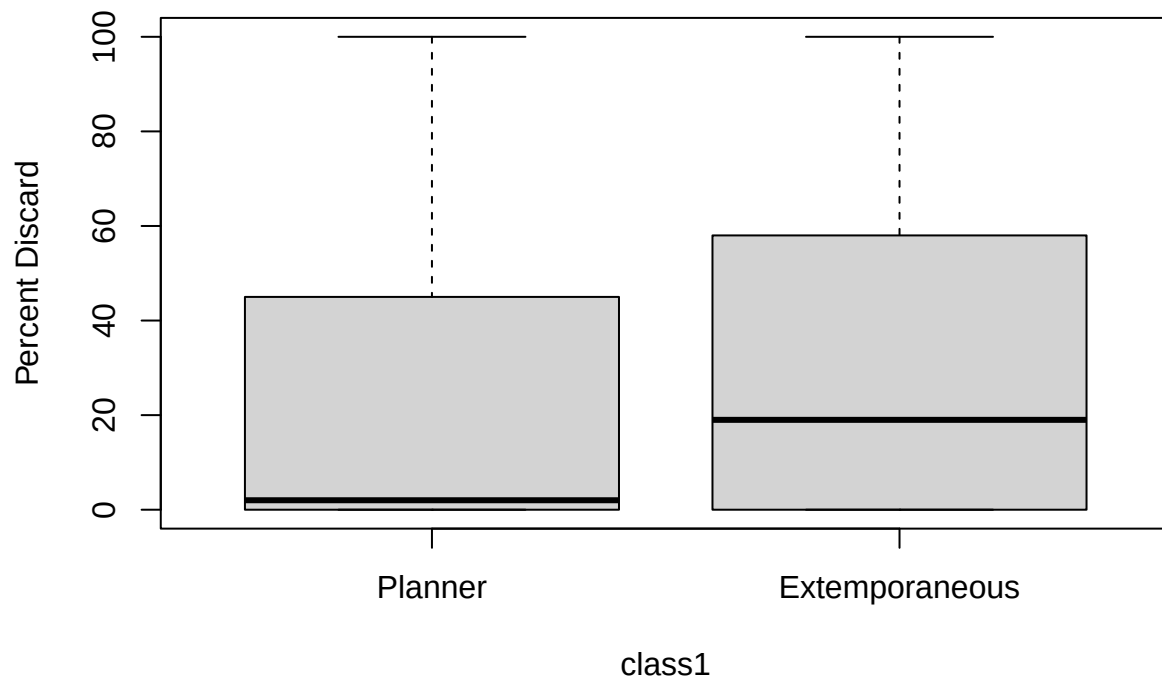
```
boxplot(percent_discard~beef,data=data2, main="Discard by Food Type", ylab="Percent Discard")
```

Discard by Food Type



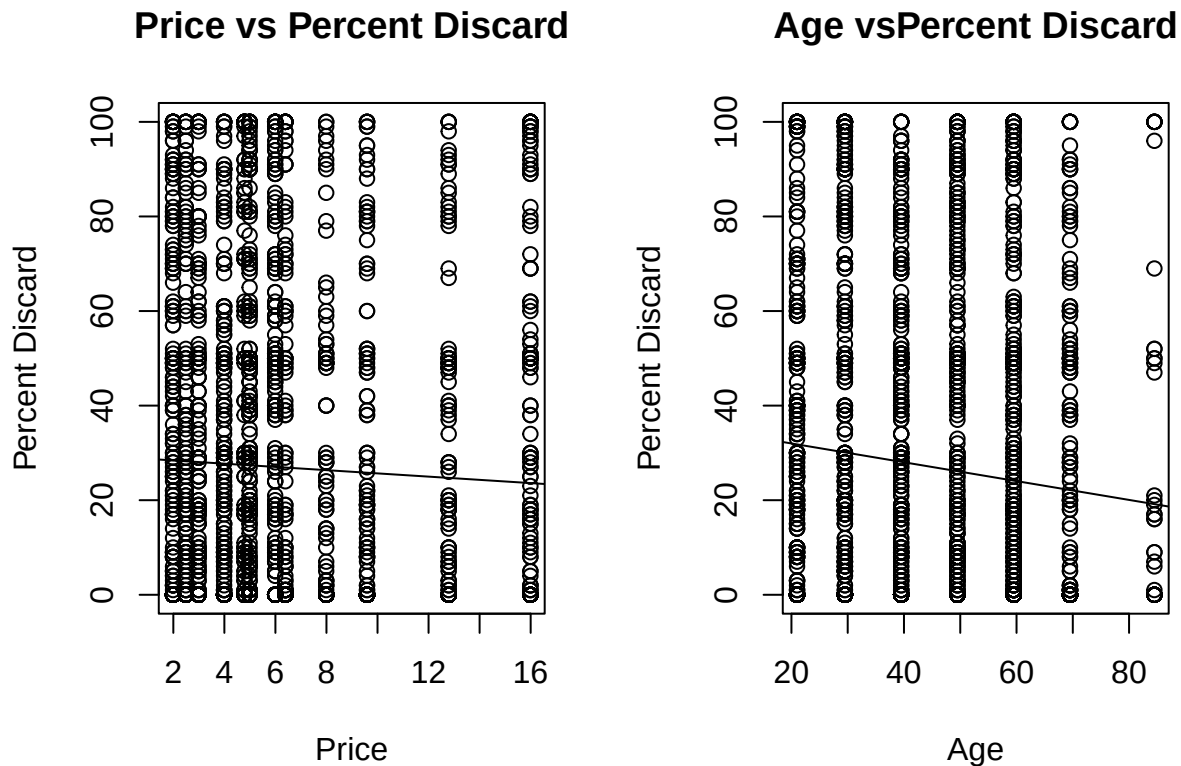
```
boxplot(percent_discard~class1, data=data2, main = "Discard by Consumer Profile", ylab="Percent Discard")
```

Discard by Consumer Profile



```
par(mfrow=c(1,2))
plot(data2$price, data2$percent_discard, main= "Price vs Percent Discard", xlab="Price", ylab="Percent D")
```

```
abline(lm(percent_discard~price, data=data2))
plot(data2$age, data2$percent_discard, main="Age vsPercent Discard", xlab="Age",ylab ="Percent Discard")
abline(lm(percent_discard~age, data = data2))
```

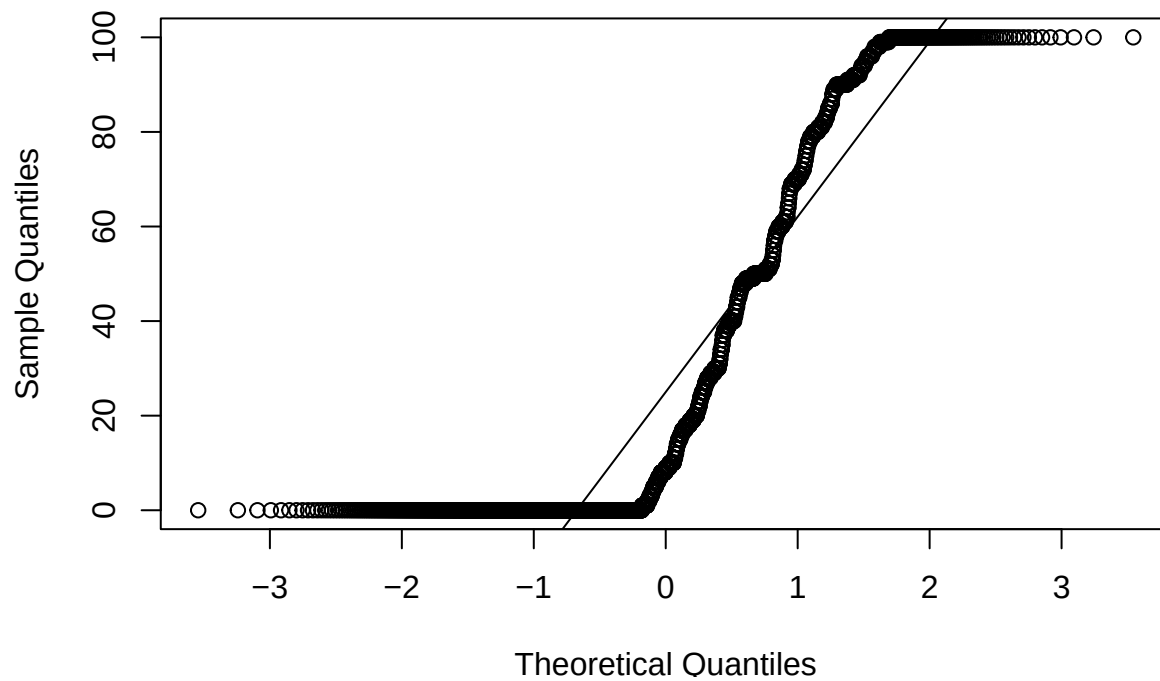


```
num_vars <- data2[, c("percent_discard", "date_days", "price", "age", "income", "hhld_size")]
cor(num_vars)
```

```
##           percent_discard    date_days      price      age
## percent_discard      1.00000000 -0.0501461810 -0.041043308 -0.094925019
## date_days           -0.05014618  1.0000000000 -0.297072402 -0.008980191
## price               -0.04104331 -0.2970724016  1.000000000  0.008125855
## age                 -0.09492502 -0.0089801907  0.008125855  1.000000000
## income              -0.09032387  0.0005610942  0.017957354  0.191998382
## hhld_size           0.01600644 -0.0165523180  0.060751803 -0.184777303
##           income    hhld_size
## percent_discard -0.0903238699  0.01600644
## date_days       0.0005610942 -0.01655232
## price           0.0179573539  0.06075180
## age             0.1919983817 -0.18477730
## income          1.0000000000  0.29926223
## hhld_size       0.2992622319  1.00000000
```

```
qqnorm(data2$percent_discard)
qqline(data2$percent_discard)
```

Normal Q-Q Plot



```
num_vars<-data2[,sapply(data2, is.numeric)]
num_vars$person_id <-NULL
corr <- cor(num_vars)
round(corr,2)
```

##	percent_discard	date_days	price	age	income	hhld_size
## percent_discard	1.00	-0.05	-0.04	-0.09	-0.09	0.02
## date_days	-0.05	1.00	-0.30	-0.01	0.00	-0.02
## price	-0.04	-0.30	1.00	0.01	0.02	0.06
## age	-0.09	-0.01	0.01	1.00	0.19	-0.18
## income	-0.09	0.00	0.02	0.19	1.00	0.30
## hhld_size	0.02	-0.02	0.06	-0.18	0.30	1.00
## preshop_1	-0.04	0.00	0.00	-0.07	-0.01	0.15
## preshop_2	-0.05	0.00	-0.01	-0.13	0.05	0.08
## preshop_3	-0.05	0.00	-0.02	-0.04	0.14	0.05
## preshop_4	-0.04	0.01	-0.01	-0.05	0.04	0.05
## instore_5	0.00	0.00	-0.03	-0.05	0.00	0.00
## instore_6	0.04	-0.01	0.04	-0.13	0.06	0.09
## instore_7	-0.04	0.03	-0.04	-0.05	0.05	-0.08
## tosspre_8	-0.06	-0.01	-0.02	-0.07	-0.03	0.01
## tosspre_9	-0.05	-0.01	-0.01	-0.04	-0.05	0.09
## tosspre_10	-0.12	0.02	-0.04	-0.01	0.02	-0.03
## tosspre_11	-0.14	0.00	-0.04	0.11	-0.08	-0.05
## tosspre_12	-0.07	0.02	-0.02	0.01	0.06	0.01
## tosspre_13	-0.05	0.01	-0.02	0.02	0.10	-0.05
## tosspre_14	-0.01	0.02	-0.04	0.13	0.04	0.02
## tosspost_15	-0.11	-0.03	0.01	0.04	-0.06	-0.02
## tosspost_16	-0.04	-0.01	-0.01	0.00	0.02	-0.01
## tosspost_17	-0.10	-0.01	-0.03	0.02	0.00	-0.03

## tosspost_18	-0.12	0.00	0.00	0.02	-0.09	-0.07
## tosspost_19	-0.12	-0.02	-0.01	0.16	0.09	0.03
## amount_20	0.07	-0.02	0.02	-0.16	0.06	0.04
## amount_21	0.17	-0.02	0.06	-0.02	0.03	0.11
## amount_22	0.15	0.00	0.01	-0.17	-0.14	0.00
## amount_23	0.05	0.00	0.01	-0.09	0.14	0.19
## amount_24	0.08	-0.02	0.06	-0.07	0.12	0.15
## amount_25	0.11	0.01	-0.02	-0.04	-0.03	0.09
## amount_26	0.08	0.01	-0.01	-0.01	0.02	0.07
## amount_27	-0.01	0.01	-0.02	0.04	0.02	0.03
## skill_28	-0.16	-0.01	0.00	0.10	0.04	-0.02
## skill_29	-0.12	0.01	0.02	0.07	0.02	-0.05
## skill_30	-0.13	0.00	-0.03	0.06	0.08	-0.03
## skill_31	-0.06	0.00	0.00	0.15	-0.03	0.00
## skill_32	-0.07	0.00	0.00	0.22	0.07	0.01
## skill_33	-0.11	-0.01	0.01	0.12	0.06	-0.13
## family_34	-0.02	-0.01	0.02	0.03	0.13	0.17
## family_35	-0.09	-0.01	-0.01	-0.05	0.13	0.10
## compost_36	-0.06	0.01	-0.01	0.11	0.15	0.01
## recycle_37	-0.10	-0.01	0.01	0.08	-0.05	0.04
##	preshop_1	preshop_2	preshop_3	preshop_4	instore_5	instore_6
## percent_discard	-0.04	-0.05	-0.05	-0.04	0.00	0.04
## date_days	0.00	0.00	0.00	0.01	0.00	-0.01
## price	0.00	-0.01	-0.02	-0.01	-0.03	0.04
## age	-0.07	-0.13	-0.04	-0.05	-0.05	-0.13
## income	-0.01	0.05	0.14	0.04	0.00	0.06
## hhld_size	0.15	0.08	0.05	0.05	0.00	0.09
## preshop_1	1.00	0.36	0.28	0.33	0.23	-0.12
## preshop_2	0.36	1.00	0.46	0.35	0.43	-0.22
## preshop_3	0.28	0.46	1.00	0.52	0.54	-0.25
## preshop_4	0.33	0.35	0.52	1.00	0.40	-0.14
## instore_5	0.23	0.43	0.54	0.40	1.00	-0.38
## instore_6	-0.12	-0.22	-0.25	-0.14	-0.38	1.00
## instore_7	0.15	0.34	0.27	0.25	0.31	-0.12
## tosspre_8	0.12	0.04	0.07	0.13	0.15	-0.22
## tosspre_9	0.04	-0.07	0.01	0.07	0.06	-0.03
## tosspre_10	-0.01	0.01	0.10	0.09	0.12	-0.07
## tosspre_11	0.07	0.08	0.01	0.04	0.11	-0.14
## tosspre_12	0.01	0.01	0.05	0.06	0.00	0.02
## tosspre_13	-0.01	0.01	0.06	0.07	-0.01	-0.14
## tosspre_14	0.09	0.09	-0.01	0.05	-0.07	-0.02
## tosspost_15	0.13	0.07	0.07	0.11	0.07	-0.15
## tosspost_16	0.06	0.05	0.07	0.17	0.17	-0.15
## tosspost_17	0.08	0.07	0.13	0.13	0.02	-0.16
## tosspost_18	0.04	0.08	0.01	0.09	0.11	-0.19
## tosspost_19	0.15	0.14	0.05	0.09	0.03	-0.07
## amount_20	-0.15	-0.17	-0.10	-0.24	-0.21	0.13
## amount_21	-0.14	-0.08	-0.12	-0.20	-0.18	0.11
## amount_22	-0.05	0.00	-0.06	-0.12	-0.08	0.08
## amount_23	-0.05	-0.02	-0.05	-0.11	-0.16	0.05
## amount_24	-0.09	-0.02	-0.04	-0.12	-0.14	0.12
## amount_25	0.06	0.05	0.00	-0.01	-0.08	-0.04
## amount_26	0.01	-0.04	-0.03	-0.04	-0.14	0.04
## amount_27	0.00	-0.09	-0.05	-0.16	-0.12	0.05

## skill_28	0.11	0.22	0.11	0.14	0.17	-0.07
## skill_29	0.07	0.13	0.00	0.09	0.11	-0.16
## skill_30	0.08	0.24	0.17	0.21	0.18	-0.07
## skill_31	0.13	0.20	0.07	0.15	0.24	-0.15
## skill_32	0.16	0.17	0.05	0.17	0.17	-0.12
## skill_33	0.06	0.09	0.08	0.15	0.17	-0.14
## family_34	0.06	0.01	-0.01	0.01	-0.14	0.21
## family_35	0.14	0.15	0.09	0.09	0.04	0.09
## compost_36	0.01	0.12	0.07	0.17	0.17	-0.13
## recycle_37	0.08	0.09	0.08	0.01	0.09	-0.04
##	instore_7	tosspre_8	tosspre_9	tosspre_10	tosspre_11	tosspre_12
## percent_discard	-0.04	-0.06	-0.05	-0.12	-0.14	-0.07
## date_days	0.03	-0.01	-0.01	0.02	0.00	0.02
## price	-0.04	-0.02	-0.01	-0.04	-0.04	-0.02
## age	-0.05	-0.07	-0.04	-0.01	0.11	0.01
## income	0.05	-0.03	-0.05	0.02	-0.08	0.06
## hhld_size	-0.08	0.01	0.09	-0.03	-0.05	0.01
## preshop_1	0.15	0.12	0.04	-0.01	0.07	0.01
## preshop_2	0.34	0.04	-0.07	0.01	0.08	0.01
## preshop_3	0.27	0.07	0.01	0.10	0.01	0.05
## preshop_4	0.25	0.13	0.07	0.09	0.04	0.06
## instore_5	0.31	0.15	0.06	0.12	0.11	0.00
## instore_6	-0.12	-0.22	-0.03	-0.07	-0.14	0.02
## instore_7	1.00	-0.03	0.00	-0.04	-0.04	-0.03
## tosspre_8	-0.03	1.00	0.38	0.30	0.26	0.07
## tosspre_9	0.00	0.38	1.00	0.23	0.21	0.07
## tosspre_10	-0.04	0.30	0.23	1.00	0.28	0.33
## tosspre_11	-0.04	0.26	0.21	0.28	1.00	0.22
## tosspre_12	-0.03	0.07	0.07	0.33	0.22	1.00
## tosspre_13	-0.01	0.18	0.11	0.31	0.36	0.38
## tosspre_14	0.02	0.03	0.15	0.16	0.22	0.36
## tosspost_15	-0.12	0.46	0.39	0.26	0.25	-0.02
## tosspost_16	0.02	0.28	0.25	0.41	0.31	0.26
## tosspost_17	0.09	0.17	0.21	0.30	0.22	0.29
## tosspost_18	-0.08	0.28	0.29	0.36	0.29	0.14
## tosspost_19	0.01	0.10	0.16	0.08	0.25	0.10
## amount_20	-0.10	-0.22	-0.24	-0.02	-0.20	0.00
## amount_21	-0.06	-0.21	-0.19	-0.16	-0.19	-0.04
## amount_22	-0.01	-0.05	-0.14	-0.07	-0.18	0.04
## amount_23	0.02	-0.09	-0.18	-0.14	-0.07	-0.02
## amount_24	-0.08	-0.19	-0.22	-0.14	-0.18	-0.01
## amount_25	0.04	-0.20	-0.12	-0.17	-0.21	-0.08
## amount_26	0.07	-0.24	-0.15	-0.13	-0.17	-0.07
## amount_27	-0.01	-0.25	-0.04	-0.12	-0.11	-0.08
## skill_28	0.09	0.11	0.09	0.09	0.24	0.08
## skill_29	-0.05	0.13	0.15	0.15	0.20	-0.04
## skill_30	0.15	0.11	0.11	0.14	0.16	0.05
## skill_31	0.10	0.25	0.24	0.15	0.21	-0.01
## skill_32	0.10	0.13	0.17	0.06	0.08	0.02
## skill_33	0.12	0.13	0.13	0.11	0.09	0.00
## family_34	-0.04	-0.13	0.03	-0.12	-0.04	-0.06
## family_35	0.02	-0.14	0.03	0.05	0.04	0.04
## compost_36	0.02	-0.01	-0.09	0.07	0.10	0.00
## recycle_37	0.02	-0.01	0.05	0.03	0.10	0.00

##	tosspre_13	tosspre_14	tosspre_15	tosspre_16	tosspre_17	
## percent_discard	-0.05	-0.01	-0.11	-0.04	-0.10	
## date_days	0.01	0.02	-0.03	-0.01	-0.01	
## price	-0.02	-0.04	0.01	-0.01	-0.03	
## age	0.02	0.13	0.04	0.00	0.02	
## income	0.10	0.04	-0.06	0.02	0.00	
## hhld_size	-0.05	0.02	-0.02	-0.01	-0.03	
## preshop_1	-0.01	0.09	0.13	0.06	0.08	
## preshop_2	0.01	0.09	0.07	0.05	0.07	
## preshop_3	0.06	-0.01	0.07	0.07	0.13	
## preshop_4	0.07	0.05	0.11	0.17	0.13	
## instore_5	-0.01	-0.07	0.07	0.17	0.02	
## instore_6	-0.14	-0.02	-0.15	-0.15	-0.16	
## instore_7	-0.01	0.02	-0.12	0.02	0.09	
## tosspre_8	0.18	0.03	0.46	0.28	0.17	
## tosspre_9	0.11	0.15	0.39	0.25	0.21	
## tosspre_10	0.31	0.16	0.26	0.41	0.30	
## tosspre_11	0.36	0.22	0.25	0.31	0.22	
## tosspre_12	0.38	0.36	-0.02	0.26	0.29	
## tosspre_13	1.00	0.35	0.11	0.24	0.19	
## tosspre_14	0.35	1.00	0.04	0.24	0.24	
## tosspost_15	0.11	0.04	1.00	0.28	0.23	
## tosspost_16	0.24	0.24	0.28	1.00	0.30	
## tosspost_17	0.19	0.24	0.23	0.30	1.00	
## tosspost_18	0.17	0.13	0.46	0.36	0.29	
## tosspost_19	0.26	0.41	0.25	0.18	0.23	
## amount_20	0.01	-0.19	-0.29	-0.14	-0.13	
## amount_21	-0.04	-0.02	-0.29	-0.10	-0.08	
## amount_22	0.00	-0.01	-0.20	-0.21	-0.13	
## amount_23	-0.09	-0.07	-0.24	-0.17	-0.14	
## amount_24	-0.12	-0.07	-0.25	-0.13	-0.20	
## amount_25	-0.07	-0.05	-0.27	-0.15	-0.10	
## amount_26	-0.05	-0.04	-0.25	-0.11	-0.09	
## amount_27	-0.05	-0.02	-0.19	-0.16	-0.17	
## skill_28	0.07	0.06	0.13	0.13	0.09	
## skill_29	0.11	0.05	0.29	0.22	0.08	
## skill_30	0.06	0.03	0.14	0.19	0.10	
## skill_31	-0.02	0.02	0.25	0.27	0.03	
## skill_32	0.05	0.09	0.13	0.14	0.06	
## skill_33	0.07	0.01	0.15	0.19	0.12	
## family_34	-0.08	-0.05	-0.02	-0.14	-0.08	
## family_35	-0.04	-0.03	0.04	-0.03	0.04	
## compost_36	0.04	0.01	0.07	0.10	0.02	
## recycle_37	-0.01	0.12	0.03	0.05	0.05	
##	tosspre_18	tosspre_19	amount_20	amount_21	amount_22	amount_23
## percent_discard	-0.12	-0.12	0.07	0.17	0.15	0.05
## date_days	0.00	-0.02	-0.02	-0.02	0.00	0.00
## price	0.00	-0.01	0.02	0.06	0.01	0.01
## age	0.02	0.16	-0.16	-0.02	-0.17	-0.09
## income	-0.09	0.09	0.06	0.03	-0.14	0.14
## hhld_size	-0.07	0.03	0.04	0.11	0.00	0.19
## preshop_1	0.04	0.15	-0.15	-0.14	-0.05	-0.05
## preshop_2	0.08	0.14	-0.17	-0.08	0.00	-0.02
## preshop_3	0.01	0.05	-0.10	-0.12	-0.06	-0.05

## preshop_4	0.09	0.09	-0.24	-0.20	-0.12	-0.11
## instore_5	0.11	0.03	-0.21	-0.18	-0.08	-0.16
## instore_6	-0.19	-0.07	0.13	0.11	0.08	0.05
## instore_7	-0.08	0.01	-0.10	-0.06	-0.01	0.02
## tosspre_8	0.28	0.10	-0.22	-0.21	-0.05	-0.09
## tosspre_9	0.29	0.16	-0.24	-0.19	-0.14	-0.18
## tosspre_10	0.36	0.08	-0.02	-0.16	-0.07	-0.14
## tosspre_11	0.29	0.25	-0.20	-0.19	-0.18	-0.07
## tosspre_12	0.14	0.10	0.00	-0.04	0.04	-0.02
## tosspre_13	0.17	0.26	0.01	-0.04	0.00	-0.09
## tosspre_14	0.13	0.41	-0.19	-0.02	-0.01	-0.07
## tosspost_15	0.46	0.25	-0.29	-0.29	-0.20	-0.24
## tosspost_16	0.36	0.18	-0.14	-0.10	-0.21	-0.17
## tosspost_17	0.29	0.23	-0.13	-0.08	-0.13	-0.14
## tosspost_18	1.00	0.15	-0.28	-0.28	-0.17	-0.21
## tosspost_19	0.15	1.00	-0.27	-0.16	-0.08	-0.13
## amount_20	-0.28	-0.27	1.00	0.47	0.39	0.30
## amount_21	-0.28	-0.16	0.47	1.00	0.46	0.33
## amount_22	-0.17	-0.08	0.39	0.46	1.00	0.28
## amount_23	-0.21	-0.13	0.30	0.33	0.28	1.00
## amount_24	-0.12	-0.19	0.33	0.44	0.27	0.43
## amount_25	-0.27	-0.13	0.29	0.39	0.43	0.20
## amount_26	-0.27	-0.12	0.33	0.44	0.38	0.26
## amount_27	-0.18	-0.07	0.30	0.28	0.32	0.20
## skill_28	0.14	0.07	-0.15	-0.20	-0.20	-0.07
## skill_29	0.24	0.11	-0.19	-0.16	-0.19	-0.12
## skill_30	0.12	0.10	-0.16	-0.24	-0.22	-0.18
## skill_31	0.22	0.09	-0.37	-0.29	-0.22	-0.15
## skill_32	0.09	0.11	-0.24	-0.15	-0.16	-0.18
## skill_33	0.15	0.06	-0.15	-0.21	-0.21	-0.15
## family_34	-0.13	-0.03	0.04	0.07	0.06	0.13
## family_35	-0.03	0.01	0.09	-0.05	0.01	0.04
## compost_36	0.10	0.10	-0.26	-0.16	-0.22	-0.19
## recycle_37	0.03	0.07	-0.18	-0.07	-0.02	-0.06
##	amount_24	amount_25	amount_26	amount_27	skill_28	skill_29
## percent_discard	0.08	0.11	0.08	-0.01	-0.16	-0.12
## date_days	-0.02	0.01	0.01	0.01	-0.01	0.01
## price	0.06	-0.02	-0.01	-0.02	0.00	0.02
## age	-0.07	-0.04	-0.01	0.04	0.10	0.07
## income	0.12	-0.03	0.02	0.02	0.04	0.02
## hhld_size	0.15	0.09	0.07	0.03	-0.02	-0.05
## preshop_1	-0.09	0.06	0.01	0.00	0.11	0.07
## preshop_2	-0.02	0.05	-0.04	-0.09	0.22	0.13
## preshop_3	-0.04	0.00	-0.03	-0.05	0.11	0.00
## preshop_4	-0.12	-0.01	-0.04	-0.16	0.14	0.09
## instore_5	-0.14	-0.08	-0.14	-0.12	0.17	0.11
## instore_6	0.12	-0.04	0.04	0.05	-0.07	-0.16
## instore_7	-0.08	0.04	0.07	-0.01	0.09	-0.05
## tosspre_8	-0.19	-0.20	-0.24	-0.25	0.11	0.13
## tosspre_9	-0.22	-0.12	-0.15	-0.04	0.09	0.15
## tosspre_10	-0.14	-0.17	-0.13	-0.12	0.09	0.15
## tosspre_11	-0.18	-0.21	-0.17	-0.11	0.24	0.20
## tosspre_12	-0.01	-0.08	-0.07	-0.08	0.08	-0.04
## tosspre_13	-0.12	-0.07	-0.05	-0.05	0.07	0.11

## tosspre_14	-0.07	-0.05	-0.04	-0.02	0.06	0.05
## tosspost_15	-0.25	-0.27	-0.25	-0.19	0.13	0.29
## tosspost_16	-0.13	-0.15	-0.11	-0.16	0.13	0.22
## tosspost_17	-0.20	-0.10	-0.09	-0.17	0.09	0.08
## tosspost_18	-0.12	-0.27	-0.27	-0.18	0.14	0.24
## tosspost_19	-0.19	-0.13	-0.12	-0.07	0.07	0.11
## amount_20	0.33	0.29	0.33	0.30	-0.15	-0.19
## amount_21	0.44	0.39	0.44	0.28	-0.20	-0.16
## amount_22	0.27	0.43	0.38	0.32	-0.20	-0.19
## amount_23	0.43	0.20	0.26	0.20	-0.07	-0.12
## amount_24	1.00	0.24	0.24	0.23	-0.11	-0.15
## amount_25	0.24	1.00	0.62	0.55	-0.04	-0.11
## amount_26	0.24	0.62	1.00	0.54	-0.07	-0.09
## amount_27	0.23	0.55	0.54	1.00	-0.07	-0.09
## skill_28	-0.11	-0.04	-0.07	-0.07	1.00	0.47
## skill_29	-0.15	-0.11	-0.09	-0.09	0.47	1.00
## skill_30	-0.20	-0.16	-0.13	-0.14	0.48	0.38
## skill_31	-0.21	-0.10	-0.19	-0.15	0.30	0.37
## skill_32	-0.15	-0.06	-0.11	-0.01	0.34	0.34
## skill_33	-0.13	-0.13	-0.16	-0.14	0.38	0.38
## family_34	0.04	0.17	0.12	0.08	0.01	-0.03
## family_35	-0.01	0.11	0.05	0.04	0.10	0.05
## compost_36	-0.11	-0.05	-0.07	-0.04	0.04	0.09
## recycle_37	-0.04	0.08	0.05	0.10	0.08	0.11
##	skill_30	skill_31	skill_32	skill_33	family_34	family_35
## percent_discard	-0.13	-0.06	-0.07	-0.11	-0.02	-0.09
## date_days	0.00	0.00	0.00	-0.01	-0.01	-0.01
## price	-0.03	0.00	0.00	0.01	0.02	-0.01
## age	0.06	0.15	0.22	0.12	0.03	-0.05
## income	0.08	-0.03	0.07	0.06	0.13	0.13
## hhld_size	-0.03	0.00	0.01	-0.13	0.17	0.10
## preshop_1	0.08	0.13	0.16	0.06	0.06	0.14
## preshop_2	0.24	0.20	0.17	0.09	0.01	0.15
## preshop_3	0.17	0.07	0.05	0.08	-0.01	0.09
## preshop_4	0.21	0.15	0.17	0.15	0.01	0.09
## instore_5	0.18	0.24	0.17	0.17	-0.14	0.04
## instore_6	-0.07	-0.15	-0.12	-0.14	0.21	0.09
## instore_7	0.15	0.10	0.10	0.12	-0.04	0.02
## tosspre_8	0.11	0.25	0.13	0.13	-0.13	-0.14
## tosspre_9	0.11	0.24	0.17	0.13	0.03	0.03
## tosspre_10	0.14	0.15	0.06	0.11	-0.12	0.05
## tosspre_11	0.16	0.21	0.08	0.09	-0.04	0.04
## tosspre_12	0.05	-0.01	0.02	0.00	-0.06	0.04
## tosspre_13	0.06	-0.02	0.05	0.07	-0.08	-0.04
## tosspre_14	0.03	0.02	0.09	0.01	-0.05	-0.03
## tosspost_15	0.14	0.25	0.13	0.15	-0.02	0.04
## tosspost_16	0.19	0.27	0.14	0.19	-0.14	-0.03
## tosspost_17	0.10	0.03	0.06	0.12	-0.08	0.04
## tosspost_18	0.12	0.22	0.09	0.15	-0.13	-0.03
## tosspost_19	0.10	0.09	0.11	0.06	-0.03	0.01
## amount_20	-0.16	-0.37	-0.24	-0.15	0.04	0.09
## amount_21	-0.24	-0.29	-0.15	-0.21	0.07	-0.05
## amount_22	-0.22	-0.22	-0.16	-0.21	0.06	0.01
## amount_23	-0.18	-0.15	-0.18	-0.15	0.13	0.04

## amount_24	-0.20	-0.21	-0.15	-0.13	0.04	-0.01
## amount_25	-0.16	-0.10	-0.06	-0.13	0.17	0.11
## amount_26	-0.13	-0.19	-0.11	-0.16	0.12	0.05
## amount_27	-0.14	-0.15	-0.01	-0.14	0.08	0.04
## skill_28	0.48	0.30	0.34	0.38	0.01	0.10
## skill_29	0.38	0.37	0.34	0.38	-0.03	0.05
## skill_30	1.00	0.28	0.35	0.35	-0.09	0.03
## skill_31	0.28	1.00	0.50	0.48	-0.07	0.03
## skill_32	0.35	0.50	1.00	0.62	0.03	0.06
## skill_33	0.35	0.48	0.62	1.00	-0.08	0.00
## family_34	-0.09	-0.07	0.03	-0.08	1.00	0.53
## family_35	0.03	0.03	0.06	0.00	0.53	1.00
## compost_36	0.12	0.11	0.05	0.12	-0.02	0.01
## recycle_37	0.02	0.11	0.12	0.09	-0.04	0.04
##	compost_36	recycle_37				
## percent_discard	-0.06	-0.10				
## date_days	0.01	-0.01				
## price	-0.01	0.01				
## age	0.11	0.08				
## income	0.15	-0.05				
## hhld_size	0.01	0.04				
## preshop_1	0.01	0.08				
## preshop_2	0.12	0.09				
## preshop_3	0.07	0.08				
## preshop_4	0.17	0.01				
## instore_5	0.17	0.09				
## instore_6	-0.13	-0.04				
## instore_7	0.02	0.02				
## tosspre_8	-0.01	-0.01				
## tosspre_9	-0.09	0.05				
## tosspre_10	0.07	0.03				
## tosspre_11	0.10	0.10				
## tosspre_12	0.00	0.00				
## tosspre_13	0.04	-0.01				
## tosspre_14	0.01	0.12				
## tosspost_15	0.07	0.03				
## tosspost_16	0.10	0.05				
## tosspost_17	0.02	0.05				
## tosspost_18	0.10	0.03				
## tosspost_19	0.10	0.07				
## amount_20	-0.26	-0.18				
## amount_21	-0.16	-0.07				
## amount_22	-0.22	-0.02				
## amount_23	-0.19	-0.06				
## amount_24	-0.11	-0.04				
## amount_25	-0.05	0.08				
## amount_26	-0.07	0.05				
## amount_27	-0.04	0.10				
## skill_28	0.04	0.08				
## skill_29	0.09	0.11				
## skill_30	0.12	0.02				
## skill_31	0.11	0.11				
## skill_32	0.05	0.12				
## skill_33	0.12	0.09				

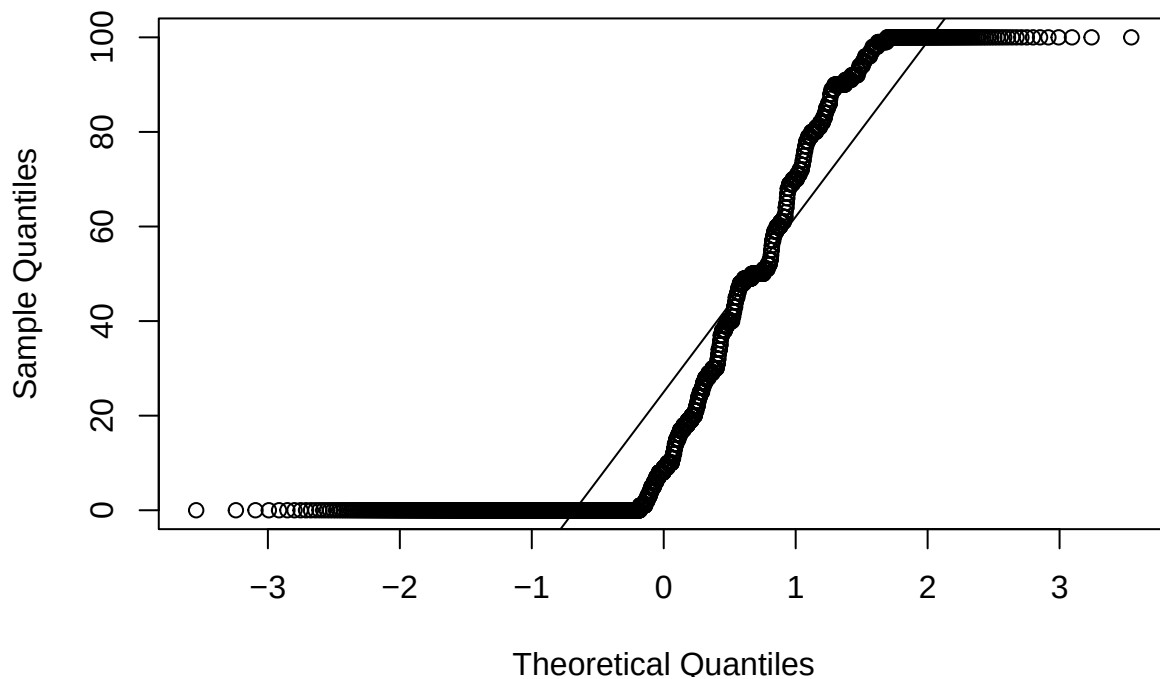
```
## family_34          -0.02      -0.04
## family_35           0.01       0.04
## compost_36          1.00       0.12
## recycle_37          0.12       1.00
```

```
corr2<-corr[, "percent_discard"]
sort(corr2)
```

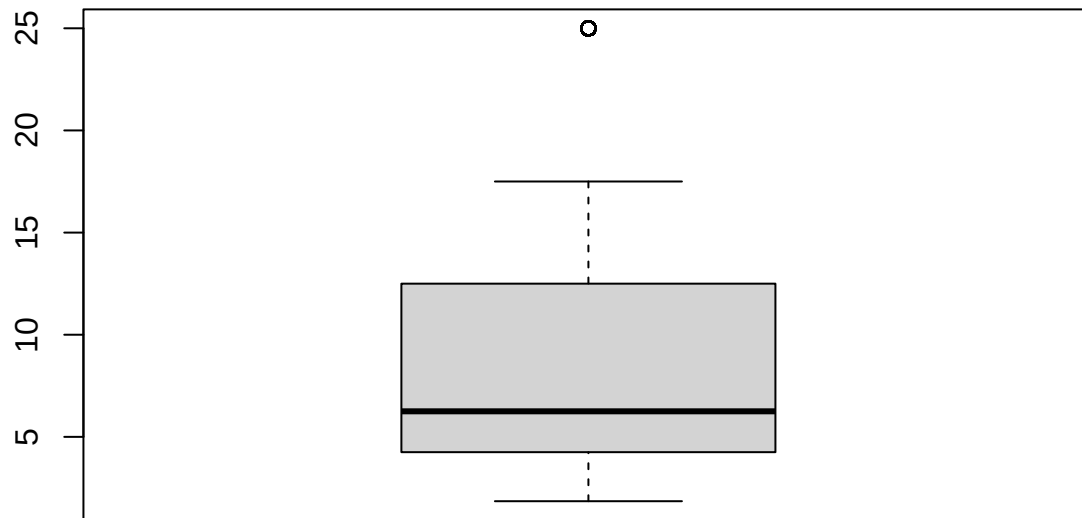
```
##      skill_28      tosspre_11      skill_30      tosspost_19      skill_29
## -0.164109540 -0.140232813 -0.133205139 -0.120594653 -0.120267255
##      tosspre_10      tosspost_18      tosspost_15      skill_33      tosspost_17
## -0.116753584 -0.116571626 -0.111850593 -0.111580074 -0.103138789
##      recycle_37      age      income      family_35      skill_32
## -0.098509800 -0.094925019 -0.090323870 -0.086570620 -0.072032074
##      tosspre_12      tosspre_8      compost_36      skill_31      preshop_2
## -0.065381851 -0.063309225 -0.061483977 -0.056995716 -0.054567291
##      tosspre_13      date_days      preshop_3      tosspre_9      preshop_4
## -0.054509841 -0.050146181 -0.047762198 -0.045124013 -0.043497559
##      preshop_1      price      tosspost_16      instore_7      family_34
## -0.042783019 -0.041043308 -0.040414885 -0.036659435 -0.018844329
##      tosspre_14      amount_27      instore_5      hhld_size      instore_6
## -0.008003652 -0.006901558 0.003849754 0.016006443 0.037495067
##      amount_23      amount_20      amount_24      amount_26      amount_25
## 0.053882635 0.070055458 0.078017209 0.081535640 0.106620715
##      amount_22      amount_21 percent_discard
## 0.151496960 0.173365302 1.000000000
```

```
qqnorm(data2$percent_discard)
qqline(data2$percent_discard)
```

Normal Q-Q Plot



```
boxplot(data2$income)
```



```
boxplot(data2$price)
```

